Implications

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Review

For propositions P and Q:

expression	name	true exactly when
$\neg P$	negation	P is false
$P \wedge Q$	conjunction	both are true
$P \vee Q$	disjunction	at least one is true

Mathematical $\vee$ is inclusive: $P \vee Q$ is true when both P and Q are true.
A **truth assignment** assigns either $\mathbf{T}$ or $\mathbf{F}$ to each propositional variable.
A truth table lists the truth value of a propositional form under every truth assignment.

Propositional forms

A **propositional form** is an expression built from propositional variables and logical connectives.
Compare

$$P \wedge (\neg P), \qquad P \vee (\neg P), \qquad P \wedge Q.$$

Predict

Which form is always false? Which is always true? Which depends on the assignment?

Classifying propositional forms

Three classifications

- A **tautology** is true under every truth assignment.
- A **contradiction** is false under every truth assignment.
- A **contingency** is true under some assignments and false under others.

P	$P \wedge \neg P$	$P \vee \neg P$	$P \wedge Q$
$\mathbf{T}$	$\mathbf{F}$	$\mathbf{T}$	depends on Q
$\mathbf{F}$	$\mathbf{F}$	$\mathbf{T}$	$\mathbf{F}$

Thus the three forms are, respectively, a contradiction, tautology, and contingency.

Logical equivalence

Logical equivalence

Two propositional forms are **logically equivalent** when they have the same truth value under every truth assignment.

We write

$$A \equiv B$$

to mean that forms A and B are logically equivalent.
The symbol $\equiv$ compares two forms; it is not the same as assigning a value to a program variable.
A truth table can establish equivalence by showing identical result columns.

Conditional statements

Let P and Q be propositions.
We often use sentences such as

- "If P, then Q."
- "Q if P."
- "P only if Q."
- "Whenever P, Q."

All four express the same logical structure: the claim that whenever P is true, Q is also true.
We denote this structure by

$$P \Rightarrow Q$$

and usually read it "P implies Q." This compound proposition is called an **implication**. Here P is the **hypothesis** and Q is the **conclusion**.

Truth table for implication

The following table is the definition of implication in classical propositional logic.

P	Q	$P \Rightarrow Q$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

The implication is false only when its hypothesis is true and its conclusion is false.

Why implication matters

An implication expresses a claim about what must be true when a condition is satisfied:

$$\text{condition} \Rightarrow \text{conclusion}.$$

This pattern appears throughout mathematics and computer science.

- A theorem often has the form assumptions $\Rightarrow$ conclusion.
- A rule states what must hold when its condition is satisfied.
- A claim about an algorithm states what its output must satisfy when the input or program state satisfies a given condition.

Vacuous truth

From the definition, $P \Rightarrow Q$ is true whenever P is false, regardless of the truth value of Q.

In this situation, the implication is said to be **vacuously true**.

Vacuous truth follows from the definition of implication. It does not mean that P causes Q or that the two propositions are related in meaning.

Equivalent form of implication

The implication $P \Rightarrow Q$ is false only when

$$P = \mathbf{T} \quad \text{and} \quad Q = \mathbf{F}.$$

Therefore, the implication holds exactly when either P is false or Q is true.

$$P \Rightarrow Q \;\equiv\; (\neg P) \vee Q.$$

Recall that $\vee$ is inclusive.

Truth-table verification

P	Q	$P \Rightarrow Q$	$\neg P$	$(\neg P) \vee Q$
T	**T**	**T**	**F**	**T**
T	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**
F	**F**	**T**	**T**	**T**

The columns for $P \Rightarrow Q$ and $(\neg P) \vee Q$ are identical.
Therefore,

$$P \Rightarrow Q \;\equiv\; (\neg P) \vee Q.$$

Different ways to state an implication

Each expression below has the symbolic form $P \Rightarrow Q$.

English expression
If P, then Q.
Q if P.
P only if Q.
When P, then Q.
Whenever P, Q.
P is sufficient for Q.
Q is necessary for P.

Only if: an example

Let n be a particular integer, and define the propositions

$$D = \text{“}n = 4k \text{ for some integer } k\text{,”}$$
$$E = \text{“}n = 2k \text{ for some integer } k\text{.”}$$

Thus, D says that n is divisible by 4, and E says that n is even. The sentence

“n is divisible by 4 only if n is even”

has the propositional form

$$D \Rightarrow E.$$

Biconditional

Biconditional

The **biconditional**

$$P \Leftrightarrow Q$$

is read "P if and only if Q" and is true exactly when P and Q have the same truth value.

"If and only if" is often shortened to **iff**.
The biconditional contains both implications:

$$P \Leftrightarrow Q \;\equiv\; (P \Rightarrow Q) \wedge (Q \Rightarrow P).$$

Truth table for the biconditional

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	**T**	**T**	**T**	**T**
T	**F**	**F**	**T**	**F**
F	**T**	**T**	**F**	**F**
F	**F**	**T**	**T**	**T**

The column for $P \Leftrightarrow Q$ is true exactly when P and Q have the same truth value.

Converse, inverse, and contrapositive

Start with the implication $P \Rightarrow Q$.

name	form	operation
original	$P \Rightarrow Q$	–
converse	$Q \Rightarrow P$	swap
inverse	$\neg P \Rightarrow \neg Q$	negate both
contrapositive	$\neg Q \Rightarrow \neg P$	swap and negate

The four propositional forms are distinct; no equivalence has been established yet.

Contrapositive equivalence

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$\neg P \Rightarrow \neg Q$	$\neg Q \Rightarrow \neg P$
T	**T**	**T**	**T**	**T**	**T**
T	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**F**	**F**	**T**
F	**F**	**T**	**T**	**T**	**T**

Compare complete columns:

$$P \Rightarrow Q \ \equiv \ \neg Q \Rightarrow \neg P.$$

Also, the converse and inverse are equivalent to each other, but not generally to the original.

Negation of an implication

We know $P \Rightarrow Q$ fails only when P is true and Q is false. Therefore

$$\neg(P \Rightarrow Q) \;\equiv\; P \wedge (\neg Q).$$

Counterexample

To refute "if P, then Q," give one case in which P is true and Q is false.

Thus, a case in which P is true and Q is false is a counterexample to $P \Rightarrow Q$.

A data-quality rule

Consider a particular data table. Define

$$V = \text{“the table passes all required validation checks,”}$$
$$T = \text{“the table enters model training.”}$$

Suppose the policy says: “The table enters training only if all required checks pass.”
Symbolic form:

$$T \Rightarrow V.$$

A policy violation has the form

$$T \wedge (\neg V):$$

the table entered training while at least one required check failed.

Logical implication is not causation

A **causal claim** says that changing one factor would change another outcome, with other relevant conditions controlled.
The formula $P \Rightarrow Q$ constrains truth values but says nothing about:

- which event happened first,
- whether P caused Q,
- how strongly two measured variables are associated.

Data-science warning

A predictive pattern, a causal claim, and a logical implication are different objects. They should not be used interchangeably.

Exercise: translation

Consider a particular experiment. Define

$$R = \text{“the experiment is reproducible,”}$$
$$S = \text{“the random seed is recorded.”}$$

Write a symbolic form

1. If the experiment is reproducible, the seed is recorded.
2. The experiment is reproducible only if the seed is recorded.
3. The experiment is reproducible if the seed is recorded.
4. The experiment is reproducible iff the seed is recorded.

Solution

Recall R: reproducible, and S: seed recorded.

1. If R, then S: $R \Rightarrow S$
2. R only if S: $R \Rightarrow S$
3. R if S: $S \Rightarrow R$
4. R iff S: $R \Leftrightarrow S$

Sentences 1 and 2 use different English patterns but have the same logical form.

Common mistakes

1. Declaring $P \Rightarrow Q$ false whenever P is false.
2. Reversing "P only if Q" into $Q \Rightarrow P$.
3. Treating the converse as equivalent to the original.
4. Negating $P \Rightarrow Q$ as $\neg P \Rightarrow \neg Q$ instead of $P \wedge \neg Q$.
5. Reading a logical implication as a causal conclusion.

Summary

- Classify a form as a tautology, contradiction, or contingency.
- Evaluate $P \Rightarrow Q$ from its truth-table definition.
- Explain why an implication with a false hypothesis is vacuously true.
- Translate "if," "only if," and "if and only if."
- Use $P \Leftrightarrow Q$ for two-directional equivalence.
- Distinguish converse, inverse, and contrapositive, and use counterexamples to refute implications.